NIRVIKA RAJENDRA

Backend Software Engineer | Data Engineering

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PROFESSIONAL SUMMARY

Python-focused Data Engineer and Analyst with 3+ years of experience building production-grade data pipelines, ETL workflows, and data-intensive backend systems. Expertise in Python automation, SQL optimization, XML/XSLT transformations, and distributed data processing. Proven ability to reduce manual effort by 70%+ through scalable pipeline design. Master's in Data Science (GPA 3.9, UMBC) with hands-on experience across the full data lifecycle ingestion, transformation, validation, storage, and visualization.

TECHNICAL SKILLS

Languages: Python, SQL, C#, JavaScript
Data Engineering: ETL/ELT Pipelines, Apache Parquet, XML/XSLT, pandas, NumPy, openpyxl, SQLAlchemy
Databases: PostgreSQL, Microsoft SQL Server, MongoDB — schema design, query optimization, indexing
Backend & APIs: FastAPI, ASP.NET Core, Node.js/Express, RESTful APIs, Microservices
Cloud & DevOps: Microsoft Azure, AWS, Docker, GitHub Actions, Azure DevOps, CI/CD Pipelines
Visualization: Streamlit, Power BI, matplotlib
Tools & Practices: Git, Jupyter, Agile/Scrum, TDD, Unit & Integration Testing, Swagger/OpenAPI, Structured Logging

PROFESSIONAL EXPERIENCE

Data Technician — Jan 2026 – Present

University of Maryland Baltimore County | Baltimore, MD

- Developed and maintained Python-based automation scripts and data pipelines processing 34,000+ records across Scopus, ETD (Electronic Thesis and Dissertation), and MDSoar (Maryland Shared Open Access Repository)/DSpace systems, reducing manual effort by 70%.
- Built ETL workflows using XML/XSLT and Parquet-based datasets, ensuring schema compliance, data transformation accuracy, and production-grade data integrity.
- Automated ETD ingestion pipelines into institutional repositories (DSpace/MDSoar), improving processing reliability and reducing manual intervention.
- Implemented unit and integration tests for data validation workflows, ensuring correctness and production stability across releases.
- Authored technical documentation for 30+ workflows, including data flow diagrams and integration specifications, improving cross-team knowledge transfer.

Software Engineer — Aug 2021 – Nov 2023

LTM (Larsen & Toubro Infotech) | Bengaluru, India

- Developed Python and FastAPI-based data ingestion services and internal APIs supporting a financial analytics platform, processing high-volume transaction data with 99.9% uptime.
- Built ETL pipelines in Python to extract, transform, and load card transaction data covering issuance, authorization, settlement, and reconciliation — into PostgreSQL, implementing billing cycle and interest accrual logic.
- Optimized 50+ PostgreSQL queries, stored procedures, and indexes using query planning and indexing strategies, improving query performance by 35–45% and reducing dashboard latency from 2.8s to 1.6s.
- Implemented structured logging and observability (Prometheus/logging frameworks) across data pipelines, improving monitoring coverage and accelerating root-cause analysis for production incidents.
- Built secure data service APIs with JWT authentication and role-based access control (RBAC), enforcing data access policies across microservices.
- Automated deployment of data services via CI/CD pipelines using Azure DevOps and GitHub Actions, enabling zero-downtime releases and reducing deployment time by 40%.
- Collaborated in Agile teams, conducted code reviews on Python and SQL work, and mentored 4 junior engineers in data pipeline best practices.

PROJECT

Flowboard: Real-Time Kanban Board [Live Demo] **2026**

- Built full-stack real-time collaboration app using React, Node.js, Express.js, and PostgreSQL with drag-and-drop functionality.
- Implemented WebSocket-based synchronization via Socket.io enabling real-time multi-user updates with sub-100ms latency.
- Integrated JWT authentication and Swagger API documentation; deployed via GitHub Actions CI/CD pipeline.

Job Application Tracker [Live Demo] **2025**

- Built MVC-style full-stack application using Node.js, Express.js, and MongoDB with GitHub OAuth authentication.
- Designed scalable schema supporting multi-status tracking, advanced filtering, and search optimization.
- Implemented automated deployment pipeline using GitHub Actions and Render.

Personal Finance Tracker [Live Demo] **2025**

- Developed secure Fast API backend with JWT authentication and PostgreSQL for financial data management.
- Containerized application with Docker and automated deployment via CI/CD pipelines.
- Built Streamlit dashboard for real-time financial visualization and analytics.

EDUCATION

Master of Professional Studies in Data Science **Jan 2024 – Dec 2025**

University of Maryland, Baltimore County | Baltimore, MD | GPA**: 3.9 / 4.0**

Bachelor of Engineering in Computer Science **Aug 2017 – Jul 2021**

Visvesvaraya Technological University | Karnataka, India | GPA**: 8.72 / 10.0**